

9.1.1/9.1.2/9.1.3/9.1.4

Hypotheses Testing

Example 1

For a certain manufacturing process the assembly times are **normally distributed** with $\mu = 300$ sec and $\sigma = 15$ sec. At some stage the production equipment is replaced. One wonders whether in the **new** situation it still holds that $\mu = 300$ sec.

One examines all this using a sample of $n = 25$ test settings. The following applies: $\bar{x} = 292$. Does the new situation differ from the original situation? Assume that there is also a **normal distribution** in the new situation, with a given $\sigma = 15$ sec, but with an **unknown** μ .

Solution

We investigate this question through two **hypotheses** (i.e. statements about the parameters of one or more populations):

$$H_0: \mu = 300$$

null hypothesis

$$H_1: \mu \neq 300$$

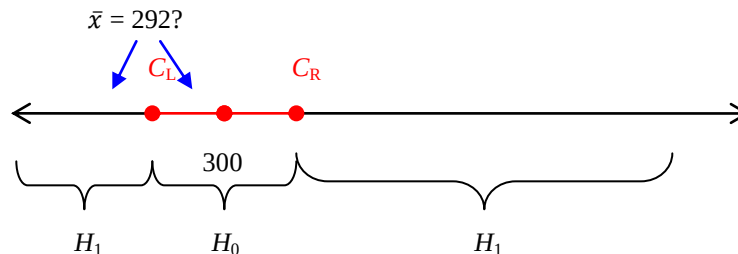
alternative hypothesis

Based on the sample result:

$$\bar{x} = 292$$

$$\text{where } \bar{X} \sim N(\mu = ?; \sigma = \frac{15}{\sqrt{25}} = 3)$$

either H_0 **or** H_1 is chosen.



We therefore determine two **critical boundaries** C_L and C_R . If $\bar{x} = 292 < C_L$, we will reject H_0 .

Determining C_L and C_R is as follows: the probability of making a **type I error** (i.e. rejecting the null hypothesis when it is true) has to be small. Therefore

$$P(H_1 | H_0) \leq \alpha, \text{ with } \alpha = 0.05 \text{ or } \alpha = 0.01$$

$$\begin{aligned} \text{So: } P(H_1 | H_0) \leq \alpha & \Leftrightarrow P(\bar{X} < C_L \text{ or } \bar{X} > C_R | \mu = 300) \leq \alpha = 0.05 \\ & \Leftrightarrow P(\bar{X} < C_L) \leq 0.025 \text{ and } P(\bar{X} > C_R) \leq 0.025 \\ & \Leftrightarrow C_L = \text{invNorm}(.025, 300, 3) = 294.12 \\ & \quad C_R = \text{invNorm}(.975, 300, 3) = 305.88 \end{aligned}$$

Since $\bar{x} = 292 \in Z = < \leftarrow ; 294.12 > \cup < 305.88 ; \rightarrow$, we reject H_0 .

Z is called the **critical region** or the **rejection region** for the test. In this example the **critical region** is the set of all values of the sample mean, where the null hypothesis is rejected.

Remarks

1. The interval [294.12; 305.88] is called the **acceptance region**. The boundaries g_L and g_R are called the **critical values**.
2. The above test is called **two-sided**. There are also **left-sided** and **right-sided** tests.

E.g. $H_0: \mu = 300$ against: $H_1: \mu < 300$ **left-sided**
 $H_0: \mu = 300$ against: $H_1: \mu > 300$ **right-sided**

A test is basically **two-sided**, unless stated otherwise.

3. The random variable $\bar{X}$ used in this test is called a **test statistic**.
4. The standardized quantity:

$$z_0 = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}} = \frac{292 - 300}{15 / \sqrt{25}} = -2.6667$$

Is called the **z-value** for the test. The test itself is called a **Z-test**. The result for this Z-test can therefore – in the case of a two-sided test – also be determined by comparing the value of z_0 with $\pm z_{\frac{1}{2}\alpha}$:

$$z_0 = -2.6667 \in Z = \{ \leftarrow ; -z_{\frac{1}{2}\alpha} = -1.96 > \cup < z_{\frac{1}{2}\alpha} = 1.96 ; \rightarrow \}$$

So H_0 will be rejected.

5. An **alternative way** to draw a conclusion is as follows:

Determine:

$$P(\bar{X} \leq 292 \mid H_0 \text{ is true}) = \text{normalcdf}(-10^{10}, 292, 300, 3) = 0.0038$$

So, when H_0 is true, it is virtually impossible that $\bar{x}$ equals 292 or less. The probability that the difference we observed between $\bar{x} = 292$ and $\mu = 300$ is based on coincidence is therefore small. So we will reject H_0 .

Rule:

$$\begin{aligned} \text{reject } H_0 &\Leftrightarrow P(\bar{X} \leq 292 \mid H_0) \leq \frac{1}{2}\alpha \\ &\Leftrightarrow 2 \cdot P(\bar{X} \leq 292 \mid H_0) \leq 2 \cdot \frac{1}{2}\alpha = \alpha \end{aligned}$$

$2 \cdot P(\bar{X} \leq 292 \mid H_0)$ is called the **P-value of the test**. It is the smallest level of significance that would lead to rejection of the null hypothesis with the given data. So the P-value of the above Z-test is equal to $2 \cdot 0.0038 = 0.0077$.

Since $p = 0.0077 \leq \alpha = 0.05$, H_0 will be rejected.

6. Using the [graphical calculator](#): **STAT → TEST → Z-Test**

7. In the case of a **left-sided** test, proceed as follows:

$$P\text{-value} = P(\bar{X} \leq 292 \mid H_0) = 0.0038 \leq \alpha = 0.05$$

So reject H_0 . Check this with the [gc](#)!

8. When formulating H_0 and H_1 , the following principles are applied:

- H_0 contains an **equality sign**.
- H_1 contains what must be **demonstrated**.

Both principles have to do with the requirement $P(H_1 \mid H_0) \leq \alpha$.

9. The type I error probability is sometimes called the [significance level](#) of the test.

10. The size of α is determined by ourselves. The maximum **type I error** is therefore fixed.

In addition to type I errors, however, there are also [type II errors](#):

$$\beta = P(H_0 \mid H_1)$$

i.e. the probability that we **fail to reject H_0 when it is false**.

The size of β depends on:

- the size of α ;
- the size of the sample.

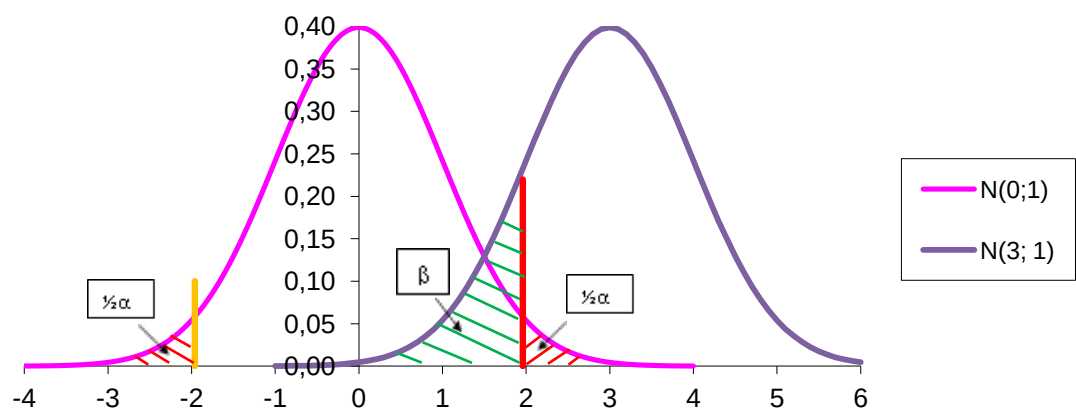


Figure 9.1 The relationship between α and β

The **smaller** α , the **larger** β . We therefore choose a **small** value for α , but not **too** small, because otherwise β becomes too large. $\alpha = 0.05$ is a [compromise value](#).

11. Certainty about the size of β is therefore not available in advance. For this reason people speak of **rejecting H_0** and of **not rejecting H_0** (so: a **strong** and a **weak** statement).
12. The expression:

$$\eta = 1 - \beta$$

is called the **power of the statistical tests**, i.e. the probability of rejecting H_0 when H_1 is true. This power must be as **high** as possible. The power of a test tells us something about the **discriminatory capacity** of the test.

The larger the distance between $\mu \in H_1$ and $H_0: \mu = 300$, the smaller β and the higher η , as the calculation below shows:

$$\begin{aligned} \mu = 301 & \Rightarrow \beta(301) = P(294.12 \leq \bar{X} \leq 305.88 \mid \mu = 301) \\ & = \text{normalcdf}(294.12, 305.88, 301, 3) \\ & = 0.9372 \end{aligned}$$

$$\mu = 305 \Rightarrow \beta(305) = 0.6152$$

$$\mu = 310 \Rightarrow \beta(310) = 0.0848$$

If $\mu \in H_1$, then:

$$\eta(\mu) = 1 - \text{normalcdf}(294.12, 305.88, \mu, 3) \quad (\text{blue color})$$

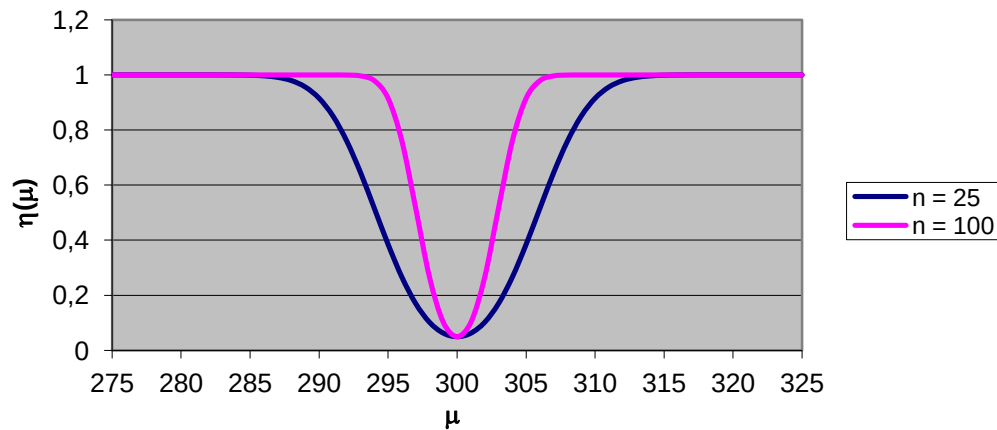


Figure 9.2 The power of a test

Check for yourself that in the case **$n = 100$** it holds that

$$Z = < \leftarrow ; 297.06 > \cup < 302.94 ; \rightarrow >$$

So

$$\eta(\mu) = 1 - \text{normalcdf}(297.06, 302.94, \mu, 1.5) \quad (\text{pink color})$$

So the **larger** the sample size, the **faster** the power gets to an acceptable height. The test therefore **discriminates** better.